

6.2 HEALTH AND DISEASE

6.2.1 Health

The term health is very frequently used by everybody. How do we define it? For a long time, **'health'** was considered a state of body and mind where there was a balance of certain 'humors' according to **'Good humor'** hypothesis. This is what early Greeks like Hippocrates as well as the Indian Ayurveda system of medicine asserted. According to this hypothesis, it was thought that persons with **'black bile'** belonged to **hot personality** and **would have fevers**. This idea was arrived at by 'a pure reflective thought'. The discovery of 'the circulation of blood by William Harvey using experimental method' and 'the demonstration of normal body temperature using thermometer in persons with black bile' disproved the 'Good humor' hypothesis.

In later years, biology stated that mind influences our immune system through neural system and endocrine system and that our immune system maintains our health. Hence, mind and mental status can affect our health. **Health** is a state of complete physical, mental and social well-being and not merely 'absence of any disease' or 'absence of physical fitness'.

Of course, our health may be affected by–

- i) **Genetic disorders** – deficiencies with which a child is born or the defects inherited by the child from its parents.
- ii) **Infections** – by either animal parasites or bacteria or virus or fungi.
- iii) **Life style** – type of food and water we take, timings of food intake, amount of rest and exercise we give to our bodies, habits such as smoking, drug and alcohol abuse, etc.

When people are healthy, they are more efficient at work. This increases productivity and brings economic prosperity. Health also increases longevity of people and reduces infant and maternal mortality. Balanced diet, personal hygiene and regular exercise are very important to maintain good health. Since times immemorial, yoga has been practiced to achieve physical and mental health. Awareness about diseases and their effect on different bodily functions, vaccination against infectious diseases, proper disposal of biological wastes, control of vectors and maintenance of hygienic food and clean water resources are necessary for achieving good health.

6.2.2 Disease

Any change from the normal state of health that causes discomfort or disability is called **disease**. When the functioning of one or more organs or systems of the body is adversely affected characterized by various signs and symptoms, we say that we are not healthy or we have a disease. Diseases can be broadly grouped into two types namely - infectious and non-infectious.

- i) **Infectious diseases:** The diseases which are easily transmitted from one person to another are called infectious diseases. These are caused by pathogens. These are very common and every one of us might have suffered from any of these diseases, at sometime or the other.
e.g. Amoebic dysentery, Malaria, Elephantiasis, Typhoid, Pneumonia, Common cold, Ringworm, etc.
- ii) **Non-infectious diseases:** The diseases which are not transmitted from one person to another and are not caused by pathogens are called non-infectious diseases.
e.g. Genetic disorders, kidney problems, heart problems, etc.

6.2.3 Common Parasites causing diseases in man

A wide range of forms that belong to protozoans, helminths, bacteria, fungi, viruses, etc., could cause diseases in man. Such disease causing forms are called pathogens. The pathogens can enter our body by various means, multiply and interfere with normal vital activities, resulting in morphological and physiological damage. A few representative members from different groups of pathogenic organisms are discussed here along with their life cycle, pathogenicity, treatment and preventive measures against the diseases caused by them.

6.2.3.1 *Entamoeba histolytica*

I. Systematic position

Phylum : Protozoa
 Subphylum : Sarcomastigophora
 Class : Rhizopodea.

Do you know the reason for the death of the famous Mughal emperor **Akbar** and the most famous Telugu King **Sri Krishna Devaraya**?

Entamoeba histolytica (Gr: *entos* - within; *amoibe* - change; *histos* - tissues; *lysis* - dissolve) is a microscopic and monogenetic parasite that inhabits the large intestine and causes **amoebic dysentery** or **amoebiasis** in man. The mode of infection is through contaminated food and drinking water. House flies, cockroaches, etc., act as 'mechanical vectors'.

It is cosmopolitan in distribution but more common in the tropical and subtropical regions of the world. It is common in the people of rural and densely populated urban areas wherever the hygienic conditions are poor.

II. Structure

Entamoeba histolytica passes through three distinct stages in its life cycle, namely:

- i) Trophozoite stage

- ii) Precystic stage and
- iii) Cystic stage

1. Trophozoite stage: It is the most active, motile, feeding and **pathogenic stage** that lives in the mucosa and sub-mucosa membranes of the large intestine. It moves with the help of a single blunt finger like pseudopodium called lobopodium which is produced in the advancing end. The body of the trophozoite is surrounded by plasmalemma. Its cytoplasm is differentiated into an outer clear, viscous, non-granular ectoplasm and the inner fluid like, granular endoplasm. Ribosomes, food vacuoles and a vesicular, **cartwheel shaped nucleus** are present in the endoplasm. However contractile vacuoles, endoplasmic reticulum, Golgi apparatus and mitochondria are absent. The absence of mitochondria indicates the '**obligate anaerobic nature**' of *Entamoeba histolytica*. It produces the proteolytic enzyme called histolysin due to which the species name '**histolytica**' was assigned to it. Due to the effect of this enzyme, the mucosa and sub-mucosa of the gut wall are dissolved releasing some amount of blood, tissue debris which are ingested by the trophozoites. Hence, the food vacuoles are with erythrocytes, fragments of epithelial cells and bacteria. The mode of nutrition is **holozoic**. Presence of '**RBC in food vacuoles**' and cartwheel shaped nucleus are the characteristic features of the trophozoites of *Entamoeba histolytica*.

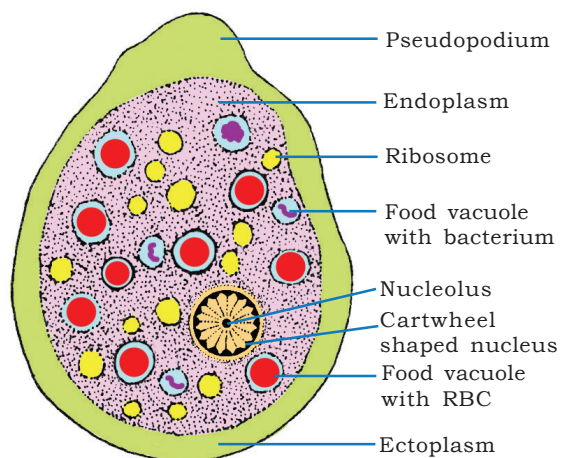


Fig. 6.1 Trophozoite Stage

2. Precystic stage: It is the non-feeding and non-pathogenic stage of *Entamoeba histolytica* that is found in the lumen of the large intestine. It is a small, spherical or oval, non-motile form. The cytoplasm of the precystic stage stores **glycogen granules** and **chromatoid bars** (made of ribonucleo protein) which act as reserve food.

3. Cystic stage: It is round in shape and is surrounded by a thin, delicate and highly resistant cyst wall. It is found in the lumen of the large intestine. The process of development of cyst wall is called **encystation** which is a means to tide over the unfavourable conditions that the parasite is going to encounter while passing to a new host. Soon after encystation, the nucleus undergoes two successive series of mitotic divisions to

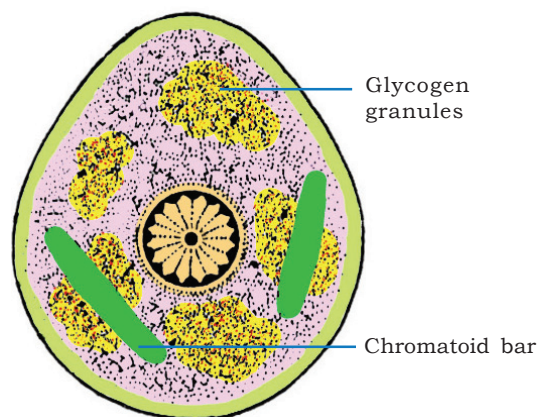


Fig. 6.2 Precystic stage

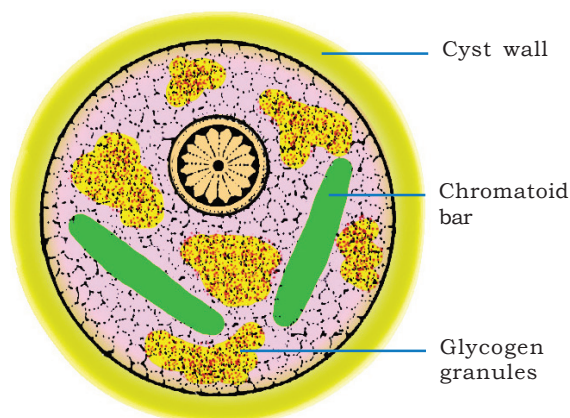


Fig. 6.3 Cystic stage

form **four daughter nuclei**. This type of cystic stage is called **tetra nucleate cyst** or **mature cyst** which is '**the stage infective to man**'.

III. Life cycle:

The trophozoites undergo binary fissions in the wall of the large intestine and produce a number of daughter entamoebae. They feed upon the bacteria and the host's tissue elements, grow in size and again multiply. After repeated binary fissions, when the trophozoites increase in number, some of the young ones enter the lumen of the large intestine and transform into **precystic stages**. Here, the precystic stages transform into **cystic stages** which

in turn develop into **tetranucleate cysts**. The entire process is completed only in a few hours. These tetranucleate cysts come out along with the faecal matter and can remain alive for about 10 days. These cysts reach new host through contaminated food and water. They pass into the small intestine of a new human host where the cyst wall gets ruptured by the action of the enzyme trypsin, releasing the **tetranucleate amoebae**. Such tetranucleate excystic amoebae are called **metacysts**.

The four nuclei of the metacyst undergo mitotic divisions and produce eight nuclei. Each nucleus gets a bit of the cytoplasm and thus eight daughter entamoebae or '**metacystic trophozoites**' are produced. These young ones develop into feeding stages called trophozoites. They invade the mucous membrane of the large intestine and grow into **mature trophozoites**.

IV. Pathogenicity

1. Intestinal amoebiasis

The trophozoites 'dissolve' the mucosal lining by **histolysin**, go deep into sub-mucosa and cause ulcers. These ulcers contain cellular debris, lymphocytes, red blood corpuscles and bacteria. It leads to the formation of abscesses in the wall of large intestine. Ultimately it results in stool with blood and mucous. This condition is called **amoebic dysentery** or **intestinal amoebiasis** or **tropical amoebiasis**.

2. Extra-intestinal amoebiasis

Some times, the trophozoites may rupture the wall of capillaries, enter the blood stream and primarily reach the liver where they may cause 'abscesses' (some call it 'secondary amoebiasis'). From there, they may go to lungs, heart, brain, kidneys, gonads, etc., and cause abscesses in those parts leading to **severe pathological conditions**.

3. Some people do not exhibit any symptoms. Such people are called '**carriers** or **asymptomatic cyst passers**' as their stool contains the tetranucleate cysts. They help in spreading the parasites to other persons.

In your view what could be the substance that forms the wall of the cyst in *Entamoeba histolytica*? Is it chitin/ protein/ lipid?

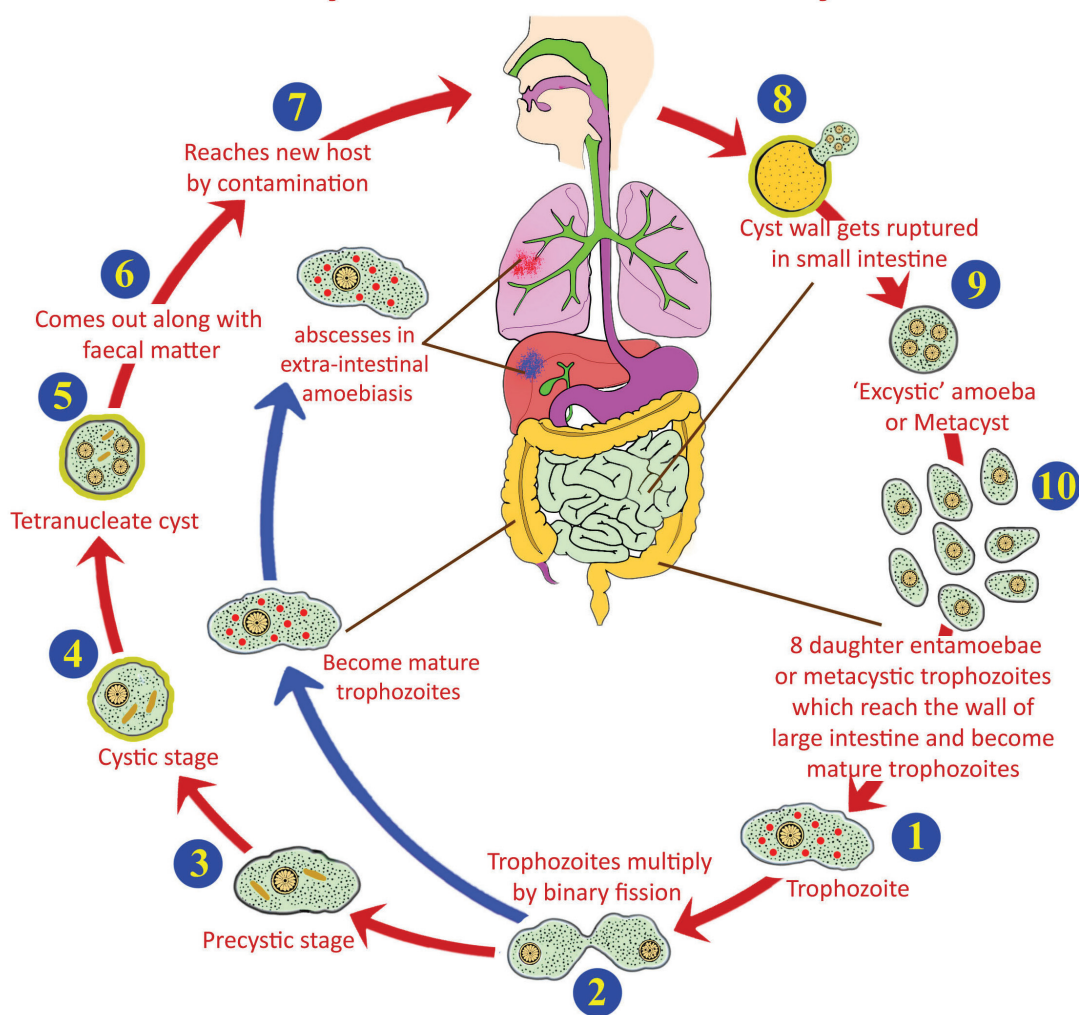


Fig. 6.4 Life cycle of *Entamoeba histolytica*

V. Prophylaxis

The following hygienic habits help prevent spread of this disease:

- Using boiled and filtered water
- Washing hands, fruits and vegetables properly
- Using septic tank toilets

In some cases, a permanent cure for amoebiasis is very hard to achieve as it frequently recurs unless the complete course of medication prescribed by a qualified medical practitioner is used.

6.2.3.2 *Plasmodium vivax*

Systematic position

Phylum	: Protozoa
Subphylum	: Apicomplexa (Sporozoa)
Class	: Telosporea
Genus	: <i>Plasmodium</i>
Species	: <i>vivax</i>

Do you know the reason for the death of '**Alexander the Great**'? No one was able to defeat **Alexander**, but the tiny Indian mosquitoes 'defeated' him. Do you know how?

Among the protozoans, *Plasmodium* is one of the most harmful parasites of man. It is a **digenetic, intracellular parasite** that lives in the **liver cells** and **RBC** of man. (It is **extracellular** in mosquito). Its **primary host** is the female **Anopheles** mosquito and the **secondary host** is man. Reservoir host is monkey. The infective stage is **sickle shaped sporozoite** and the mode of infection is **inoculation**.

Four species of *Plasmodium* cause four types of malaria in man. They are

- i) *Plasmodium vivax* – benign tertian malaria
- ii) *Plasmodium falciparum* – malignant tertian malaria
- iii) *Plasmodium ovale* – mild tertian malaria
- iv) *Plasmodium malariae* – quartan malaria

Of all these four species, *Plasmodium vivax* is the most common and most widely distributed malaria parasite. Hence its life cycle along with pathogenicity, treatment and preventive measures are briefly described here.

I. History of Malaria

The term '**Malaria**' is taken from Italian language which means 'bad air' (*Mala* - bad; *aria* - air) as it was thought that malaria was due to foul air. **Charles Laveran**, a French military doctor discovered *Plasmodium* in the blood of a malaria patient. **Sir Patrick Manson**, a Scottish doctor while working on elephantiasis, suggested that malaria might be spread by mosquitoes. **Sir Ronald Ross**, a British army doctor, while working in Secunderabad, A.P., India, identified the oocysts of *Plasmodium* in the stomach wall of female *Anopheles* mosquito on the 20th of August. In recognition of this work, Ross was awarded the **Nobel Prize** in 1902 and 20th August of every year is celebrated as **World Mosquito Day**. April 25th is the World Malaria day by WHO (from 2007). **G.B. Grassi** and others have described the life cycle of *Plasmodium vivax* in the female *Anopheles* mosquito.

II. Structure of sporozoite

The ultra-structure of the sporozoite of *Plasmodium vivax* was studied by **Garnham**. It is sickle shaped with a swollen middle part and pointed at both ends of its body. It measures about 15 microns in length and one micron in width. The body is covered by an elastic pellicle with '**microtubules**' which help in the wriggling movements of the sporozoite. The cytoplasm contains cell organelles such as Golgi complex, endoplasmic reticulum, mitochondria and a nucleus. Cytoplasm also shows many convoluted tubules of unknown function throughout the length of the body. It contains a cup like depression called **apical cup** at the anterior end into which a pair of **secretory organelles** opens. They secrete a cytolytic enzyme, which helps in the penetration of sporozoite into the liver cells.

The life cycle of *Plasmodium* is completed in two hosts as mentioned earlier.

III. Life cycle of *Plasmodium* in man (The human phase)

In man, the *Plasmodium* reproduces by asexual reproduction called **schizogony**. It occurs in liver cells (hepatocytes) as well as in RBC. In liver cells, it is called **hepatic schizogony** and in RBC it is called **erythrocytic schizogony**.

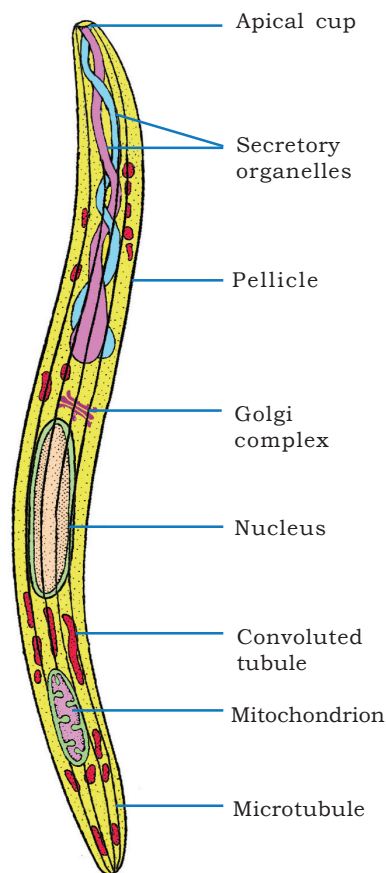


Fig. 6.5 Structure of sporozoite

1. Hepatic schizogony

This was discovered by **Shortt** and **Garnham**. Whenever, a mosquito infected by *Plasmodium* bites a man, nearly 2000 sporozoites are released into the blood of man through its saliva. Within half an hour, they reach the hepatocytes where they undergo **pre-erythrocytic** and **exo-erythrocytic** cycles.

A. Pre-erythrocytic cycle

Whenever the sporozoites reach the liver cells, they transform into **trophozoites**. They feed on the contents of the hepatic cells, assume spherical shape and attain the maximum size. This stage is called **schizont** stage. Its nucleus divides several times mitotically, followed by the cytoplasmic divisions resulting in approximately 12,000 daughter individuals called **cryptozoites** or the **1st generation merozoites**. They enter the sinusoids of the liver by rupturing the cell membrane of the schizont and the liver cells. This entire process is completed approximately in 8 days. Now these first generation merozoites have two options i.e. they can enter either fresh liver cells and continue exo-erythrocytic cycle or they can enter RBC and continue erythrocytic cycle.

Have you ever given a sample of blood for blood test? Did you observe how doctor takes the sample of your blood? Do you find any similarity between that and the bite of mosquito?

B. Exo-erythrocytic cycle

If the cryptozoites enter the fresh liver cells, they undergo the changes similar to that of the pre-erythrocytic cycle and produce the second generation merozoites called **metacryptozoites**. These are of two types- the smaller **micro-metacryptozoites** and larger **macro-metacryptozoites**. This entire process is completed approximately in two days. The macro-metacryptozoites attack fresh liver cells and continue another exo-erythrocytic cycle, whereas the micro-metacryptozoites always enter blood stream and attack fresh RBC to continue repeated erythrocytic cycles producing merozoites.

Prepatent period

The interval between 'the first entry of *Plasmodium* into the blood in the form of sporozoites and the second entry of *Plasmodium* into the blood in the form of cryptozoites is called **prepatent period**. It lasts approximately 8 days. During this period, the host does not show any clinical symptoms of the disease. It is only a means of multiplication.

2. Erythrocytic Schizogony

A. Cycle of Golgi

It was first described by **Camillo Golgi**. Hence it is also called '**Golgi cycle**'. This cycle is initiated either by the cryptozoites of pre-erythrocytic cycle or the micro-metacryptozoites of exo-erythrocytic cycle. In the fresh RBC, these stages assume spherical shape and transform into **trophozoites**. It develops a small vacuole which gradually enlarges in size, pushing the cytoplasm and nucleus to the periphery. Now the *Plasmodium* looks like a **finger ring**. Hence this stage is called **signet ring stage**. Soon it loses the vacuole, develops pseudopodia and becomes **amoeboid stage** (late trophozoite stage). With the help of pseudopodia, it actively feeds on the contents of the RBC and increases in size. As a result, the RBC grows almost double the size. This process is called **hypertrophy**. The malaria parasite digests the globin part of the ingested haemoglobin and converts the soluble haem into an insoluble crystalline **haemozoin**. It is called the '**malaria pigment**' which is a disposable product. During this stage, small red coloured dots appear in the cytoplasm of the RBC known as **Schuffner's dots**. These are believed to be the antigens released by the parasite. Now the *Plasmodium* loses the pseudopodia, increases further in size, occupies the entire RBC and becomes a schizont. It undergoes schizogony similar to that of the pre-erythrocytic cycle and produces 12 to 24 erythrocytic merozoites. They are arranged irregularly in the RBC form. Finally the erythrocyte bursts and releases the merozoites along with haemozoin into the blood. This cycle is completed approximately in 48 hours.

Incubation Period

The period between ‘the entry of *Plasmodium* into the blood in the form of sporozoite and the first appearance of symptoms of malaria in man’ is called **incubation period**. It is approximately 10 to 14 days.

B. Formation of gametocytes

After repeated cycles of erythrocytic schizogony, when the number of fresh RBC decreases, some merozoites enter the RBC and transform into gametocytes instead of continuing the erythrocytic cycle. This process generally takes place when the RBCs are present in **spleen** and **bone marrow**.

The gametocytes are of two types namely, smaller **microgametocytes** or **male gametocytes** and larger **macrogametocytes** or **female gametocytes**. The gametocytes cannot undergo further development in man as the temperature and pH of the blood of man are not suitable for further development. These gametocytes reach the blood circulation and wait to reach the next host. They degenerate and die if they are not transferred to mosquito within a week.

IV. Life cycle of *Plasmodium* in mosquito (The mosquito phase) – Ross cycle

When a female *Anopheles* mosquito bites and sucks the blood of a malaria patient, the gametocytes along with the other stages of the erythrocytic cycle reach the crop of mosquito. Here all the stages are digested except the gametocytes. Further part of the life cycle consists of:

- i) Gametogony
 - ii) Fertilization
 - iii) Formation of Ookinete & Oocysts
 - iv) Sporogony
- i) **Gametogony:** The formation of male and female gametes from the gametocytes is called **gametogony**. It occurs in the lumen of the crop of mosquito.

a. Formation of male gametes: During this process, the nucleus of microgametocyte divides into eight daughter nuclei called **pronuclei** which reach the periphery. The cytoplasm is pushed out in the form of eight flagella like processes. Into each flagellum like process, one pronucleus enters and forms a **micro gamete** or **male gamete**. These male gametes show lashing movements like those of flagella and get separated from the cytoplasm of microgametocyte. This process is called **exflagellation**.

b. Formation of female gamete: The female gametocyte undergoes a few changes and transforms into a female gamete. This process is called **maturation**. The nucleus of the female gamete moves towards the periphery and the cytoplasm at that point forms a projection. This projected region is called the **fertilization cone**.

Do you know: Why a minimum of ten days is required for the occurrence of the first symptoms of malaria in man after the entry in the form of sporozoites?

SEXUAL LIFE CYCLE IN MOSQUITO

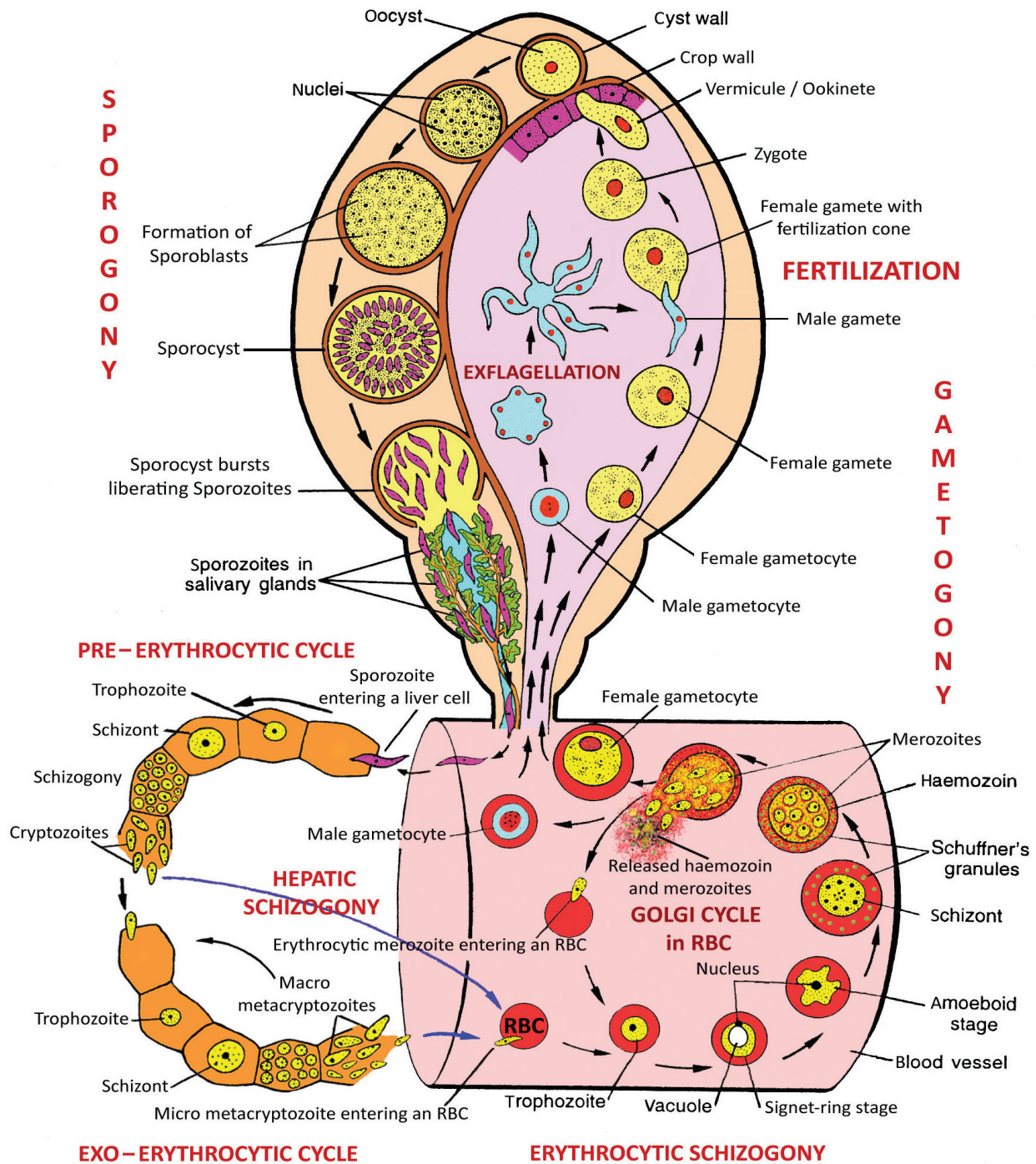


Fig. 6.6 Life cycle of *Plasmodium vivax*

- ii) **Fertilization:** The fusion of male and female gametes is called fertilization. It also occurs in the lumen of the crop of the mosquito. When an actively moving male gamete comes into contact with the fertilization cone of the female gamete, it enters it. The pronuclei and cytoplasm of these two gametes fuse with each other, resulting in the formation of a synkaryon. Since the two gametes are dissimilar in size, this process is known as **anisogamy**. The female gamete that bears the synkaryon is called the **zygote** which is round and non-motile.
- iii) **Formation of ookinete and oocysts:** The zygote remains inactive for some time and then transforms into a long, slender, motile, vermiform **ookinete** or **vermicule** within 18 to 24 hours. It pierces the wall of the crop and settles beneath the **basement membrane**. It becomes round and secretes a cyst around its body. This encysted ookinete is now called **oocyst**. About 50 to 500 oocysts are formed on the wall of the crop and appear in the form of small nodules. (Sir Ronald Ross identified these oocysts for the first time).
- iv) **Sporogony:** The formation of sporozoites in the oocysts is called **sporogony**. According to Bano, the nucleus of the oocyst first undergoes reduction division followed by repeated mitotic divisions resulting in the formation of sporoblasts. The nucleus of the sporoblast divides and forms several nuclei. Each nucleus is surrounded by a little bit of the cytoplasm and transforms into a sickle shaped sporozoite. Oocyst with such sporozoites is called **sporocyst**. When this sporocyst ruptures, about 10,000 sporozoites are liberated into the haemocoel of the mosquito. From there, they travel into the salivary glands and are ready for infection. The life cycle of *Plasmodium* in mosquito is completed in about 10 to 24 days.

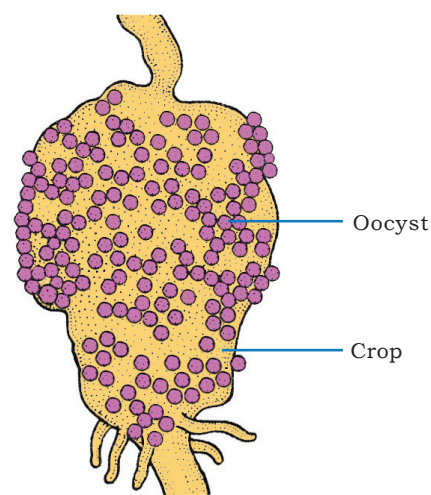


Fig. 6.7 Oocysts of *Plasmodium* on the crop of female *Anopheles*

Plasmodium exhibits alternation of generations in its life cycle. Can this statement be justified? How?

V. Pathogenicity

Plasmodium vivax causes benign tertian malaria. The clinical symptoms of this disease include bouts of fever which can be expressed in three stages namely cold stage, hot stage and sweating stage.

Cold stage: Chills, shivering, headache and giddiness.

Hot stage: High temperature of the body, rapid breathing and an increase in the pulse rate.

Sweating stage: Profuse sweating followed by lowering of the body temperature to the normal level.

Anaemia and **splenomegaly** are noticed in a chronic malaria patient. Relapse of malaria may also be noticed some times.

Why does the spleen enlarge in the case of chronic malaria patient?

Ans: In case of repeated infections, spleen shows an abnormal immune response and enlarges. (Tropical splenomegaly syndrome)

Relapse of malaria: Some of the stages of macro-metacryptozoites may survive for a long period in liver as dormant stages called **hypnozoites**. Reactivation of these hypnozoites leads to the initiation of fresh erythrocytic cycles resulting in the new attacks of malaria. This is referred to as **relapse of malaria**.

There are certain people who are immune to malaria. Do you know what the reason is for immunity against malaria ? (Ask your teacher to explain immunity to malaria in people heterozygous for sickle cell anaemia).

VI. Treatment

Malaria can be cured by **quinine** which is an alkaloid extracted from the bark of *Cinchona officinalis*.

VII. Prophylaxis

- ❖ Spraying of DDT, BHC, etc., insecticides at intervals in mosquito breeding places like pools, ponds, ditches and stagnant water.
- ❖ Introduction of larvivorous fishes like *Gambusia*, insectivorous plants like *Utricularia* into the places where mosquitoes breed.
- ❖ Avoiding mosquito bite by using mosquito nets, mosquito repellents, etc.
- ❖ Spraying of kerosene, pyrethrum oil, etc., on stagnant water.

6.2.3.3 *Ascaris lumbricoides*

Do you know the reason for giving the species name '*lumbricoides*' for *Ascaris*?

Clue : There is an earthworm called *Lumbricus*

Phylum : Nematoda

Class : Phasmidia

Ascaris lumbricoides is commonly called the **common round worm**. It is one of the most common entericparasites of man. It is monogenetic, dimorphic, pseudocoelomate parasite that lives in the small intestine of man, more frequently in children. It is cosmopolitan in distribution. Mode of infection is through contaminated food and water. Infective stage is the embryonated egg with the **2nd stage rhabditiform larva**.

I. Structure: Sexes are separate and the sexual dimorphism is distinct. In both males and females, the body is elongated and cylindrical. Mouth is present at the extreme anterior end and is surrounded by three chitinous lips. Close to the mouth mid ventrally, there is a small aperture called excretory pore.

1. Male: It has a curved posterior end which is considered the tail. The posterior end possesses a cloacal aperture and a pair of equal sized chitinous '**pineal spicules**' or '**pineal setae**' which serve to transfer the sperms during copulation.

2. Female: It has a straight posterior end, the tail. The female genital pore or vulva is present mid ventrally at about one third the length from mouth. The anus is present a little in front of the tail end.

II. Life history

Copulation takes place in the small intestine of man. After copulation, the female releases approximately two lakh eggs per day. Each egg is surrounded by '**a protein coat**' with rippled surface. Hence the eggs of *Ascaris* are described as '**mammillated eggs**'. The protein coat is followed by a chitinous shell and a lipid layer internally. These eggs are passed out along with faecal matter. In the moist soil, development takes place inside the egg leading to the formation of the **1st stage rhabditiform larva** is produced. It undergoes the **1st moulting** and becomes the **2nd stage rhabditiform larva** which is the stage '**infective to man**'. They reach the alimentary canal of man through contaminated food and water.

Do you know: 'If the conditions are favourable, the embryonated eggs of *Ascaris* can remain viable for about 6 years in moist soil'. Can you guess the reason for this?

In the small intestine, the shell gets dissolved and the **2nd stage larva** is released. Now it undergoes **extra intestinal migration**. First it reaches the liver through the hepatic portal vein. From there it reaches the heart through the post caval vein. It goes to the lungs through the pulmonary arteries. In the alveoli of the lungs it undergoes the **2nd moulting** to produce the **3rd stage larva**. It undergoes the **3rd moulting** so that the **4th stage larva** is produced in the alveoli only. It leaves the alveoli and reaches the small intestine again, through the bronchi, trachea, larynx, glottis, pharynx, oesophagus and stomach. In the small intestine, it undergoes the

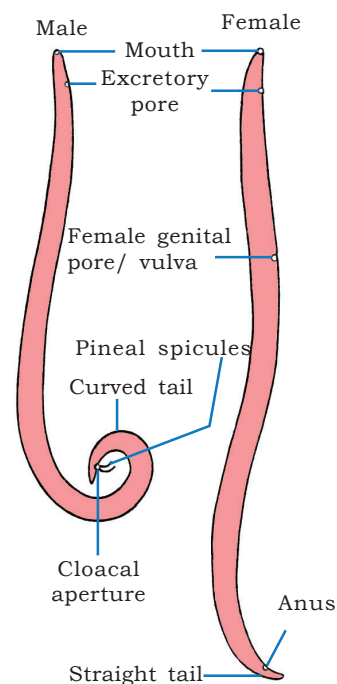


Fig. 6.8 *Ascaris lumbricoides*

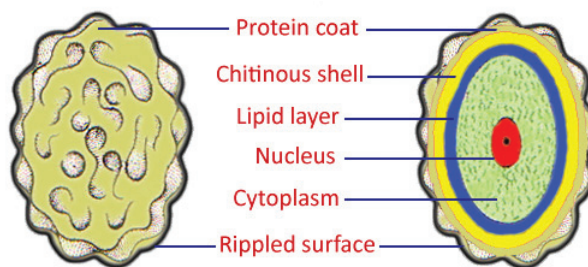


Fig. 6.9 Mammillated eggs of *Ascaris*

4th moulting (final moulting) to become a young one which attains sexual maturity within 8 to 10 weeks.

III. Pathogenicity

The disease caused by *Ascaris lumbricoides* is called '**ascariasis**'. The disease is asymptomatic if the number of worms is less. A heavy infection can cause nutritional deficiency and severe abdominal pain. It also causes stunted growth in children.

Try to find out the reason for stunted growth noticed in children due to *Ascaris lumbricoides* infection. (It is believed to produce certain substances which inhibit the Human Growth Hormone)

IV. Prophylaxis: Same as that of *Entamoeba*.

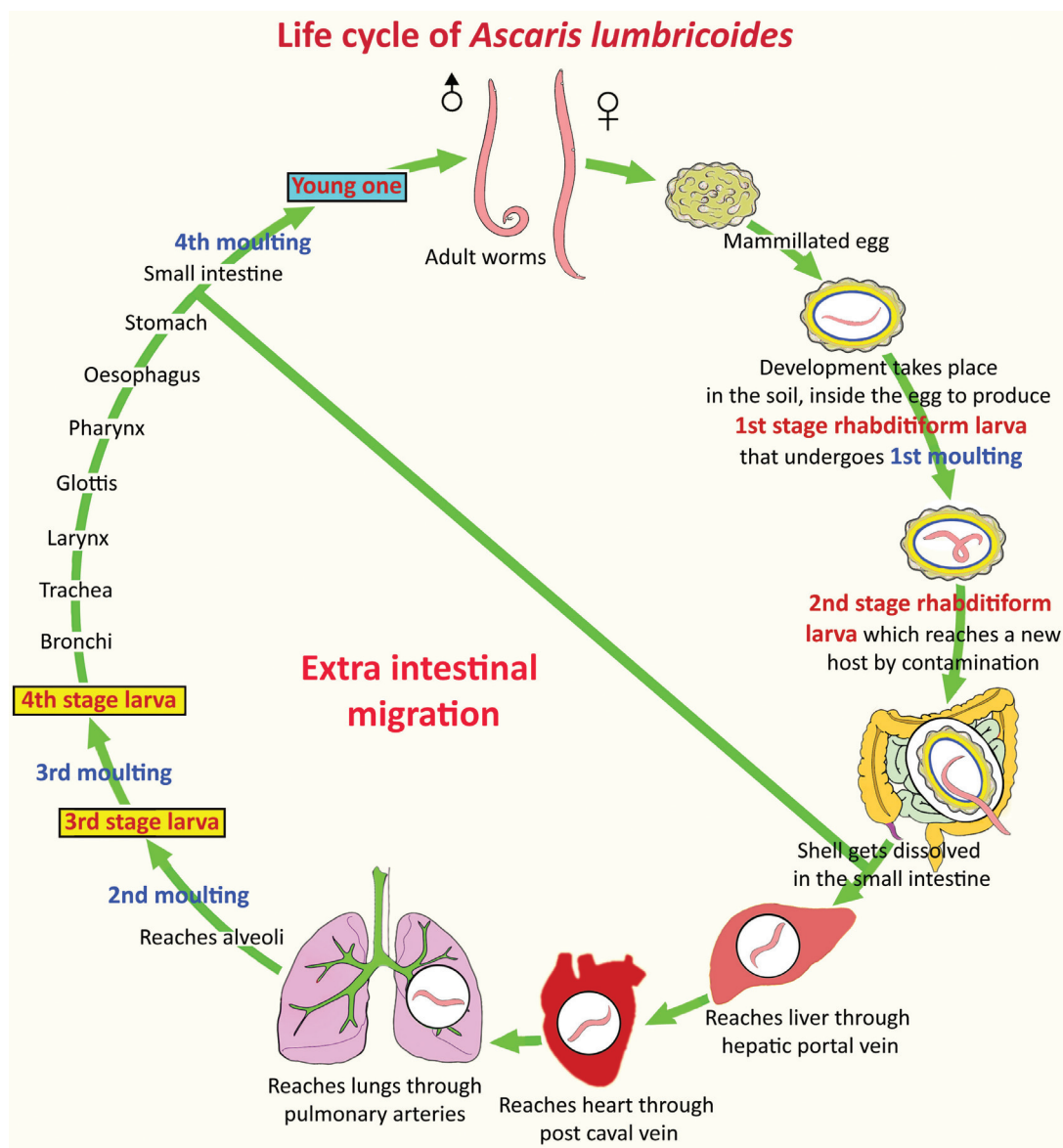


Fig. 6.10 Life cycle of *Ascaris lumbricoides*

6.2.3.4 *Wuchereria bancrofti*

Phylum : Nematoda

Class : Phasmdia

Wuchereria bancrofti is commonly called the filarial worm as it causes filariasis in human beings. It is a digenetic, dimorphic, pseudocoelomate and histozoic parasite that lives in the lymph vessels of man. **Sir Patrick Manson** identified **female Culex** mosquito as its secondary host.

I. Structure

Sexes are separate and the sexual dimorphism is distinct. The body is long, narrow, filiform and creamy-white in colour. The anterior end is blunt and the posterior end is pointed. Mouth is present at the anterior end and is without any lips.

1. Male worm: Its posterior end is curved with a cloacal aperture. A pair of unequal, chitinous '**pineal spicules**' or '**copulatory spicules**' is present in the cloacal region.

2. Female worm: Its posterior end is straight. Anus is present near the posterior end. The female genital pore or vulva is present mid ventrally at about one third the length from the mouth. It is **ovoviviparous**.

II. Life cycle

As we already discussed, it completes its life cycle in two hosts namely man and female *Culex* mosquito.

1. In man

Both male and female worms are found coiled together in the lymphatic vessels of man. After copulation the female releases the sheathed microfilaria larvae into the lymph of man. Each sheathed microfilaria larva measures 0.2 to 0.3 mm in length. It is surrounded by a loose cuticular sheath which is supposed to be the modified shell. They migrate to the blood circulation and reside in the deeper blood vessels during the day time. They move to the peripheral blood circulation during the night time between 10.00 pm and 4.00 am. This tendency is referred to as '**nocturnal periodicity**'. When a female *Culex* mosquito sucks the blood of an infected person, they enter the gut of mosquito. They die if they are not transferred to mosquito **within 70 days**.

Could you guess the reason for nocturnal periodicity of sheathed microfilaria?

2. In mosquito

In the midgut of mosquito, the sheath of the larva is dissolved within 2 to 6 hours of the infection. The ex-sheathed microfilaria larva penetrates the gut wall



Fig. 6.11 A case of elephantiasis

and reaches the haemocoel of mosquito. From there, it reaches the '**thoracic muscles**' and transforms into a '**sausage shaped larva**' within two days. It is called the **first stage larva** or **first stage microfilaria**. This undergoes two moultings within 10 to 20 days and transforms into a long, **infective '3rd stage microfilaria**'. It reaches the labium of the mosquito.

Why can't we consider the sheathed microfilaria larva the 1st stage microfilaria though it is the first stage produced in the life cycle of *Wuchereria*?

Note : A larva is a self supporting immature young one. Until the embryo can feed on its own, it can not be considered a larva.

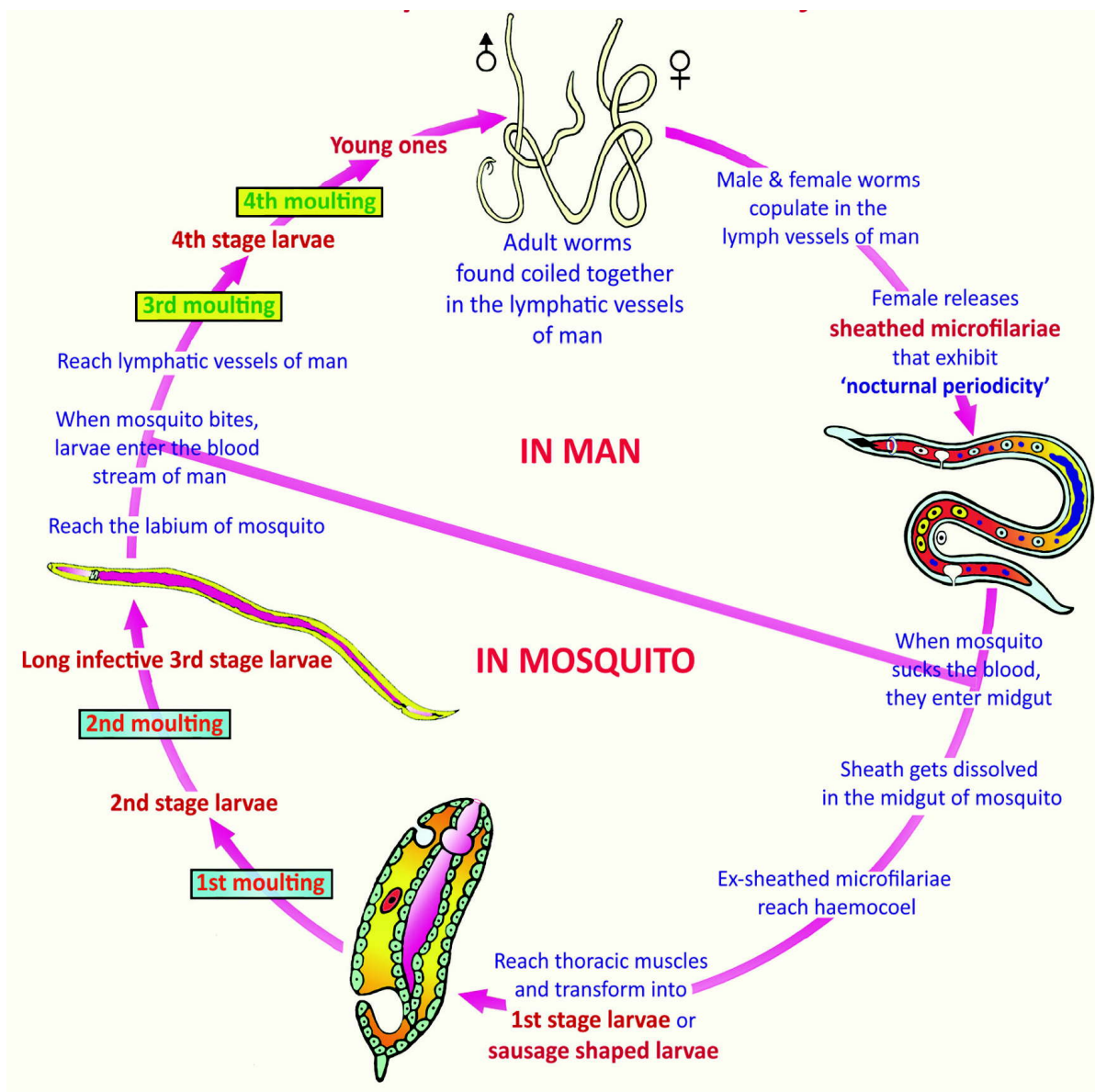


Fig. 6.12 Life cycle of *Wuchereria bancrofti*